

To appear in ICML 2026

OCNR: Stabilizing Self-Play by Mitigating Iteration-Collapse with One-Class Novelty Rewards

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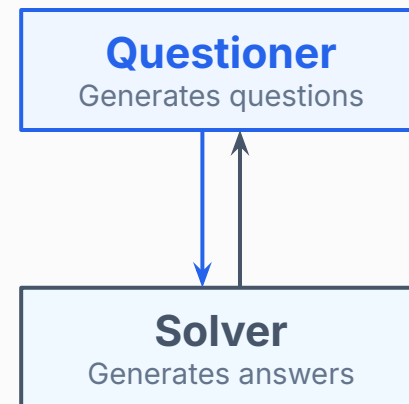
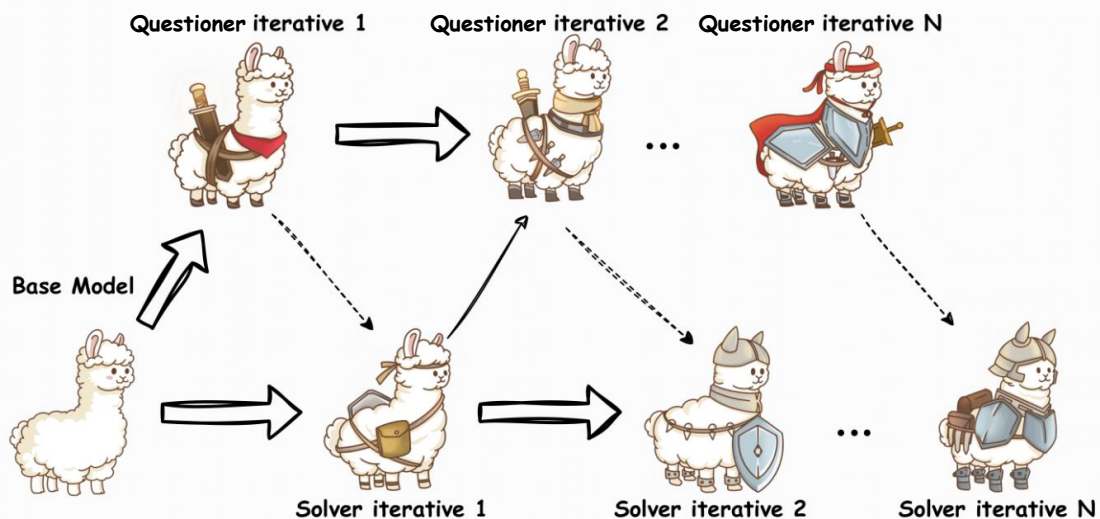
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Background

Post-Training via Closed-Loop Self-Play

Self-Play pipelines offer a path toward open-ended improvement, in principle.

Iteration Collapse: Performance initially improves but subsequently degrades.

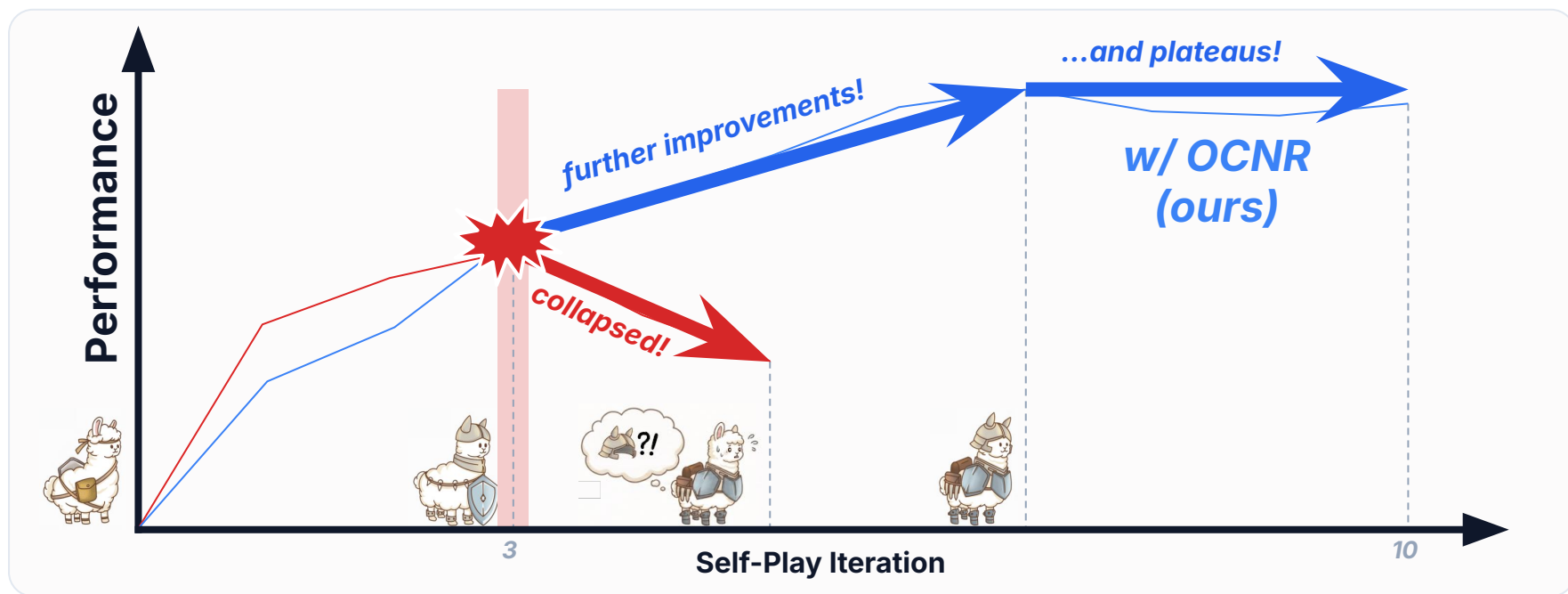


Background

The Wall: Iteration Collapse

Self-Play pipelines offer a path toward open-ended improvement, in principle.

Iteration Collapse: Performance initially improves but subsequently degrades.



Approach

Novelty: Beyond Diversity

(1) Validity

Questions should be solvable and well-formatted.

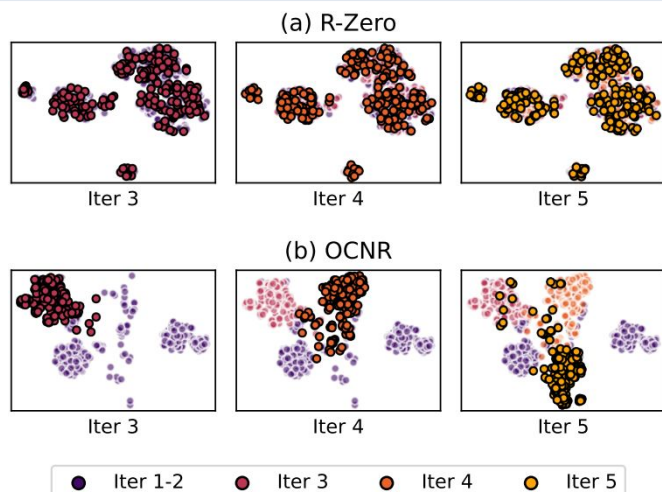
(2) Difficulty

Questions should be tailored to the solver's level.

CRITICAL

(3) Diversity

Questions should encompass a wide range of types.



✓ Diverse within each iteration,
⚠ but, non-diverse across iterations!

✓ Diverse within each iteration,
✓ and diverse across iterations!

2D t-SNE of question embeddings across self-play iteration

Problems must be genuinely new from the solver's perspective.

→ Not just diversity, but novelty!

Approach

OCNR: One-Class Novelty Reward

Seen Detector (via **Learnable Prototype Vector**)

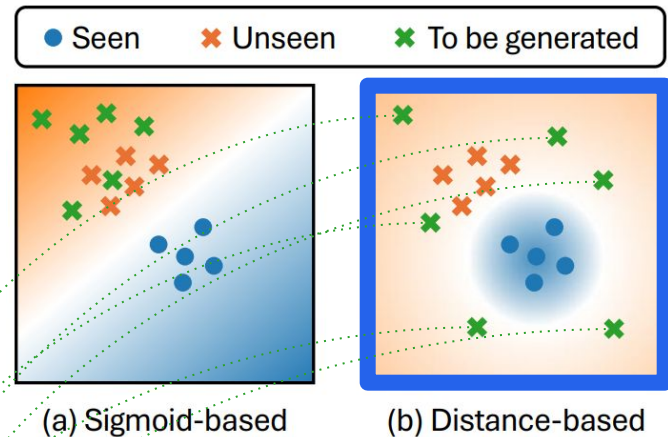
$$\text{SEEN}(q_t) = \exp \left(-\frac{\|h(q_t) - w_s\|^2}{\tau} \right)$$

One-Class Novelty Reward (OCNR)

$$R_Q^{\text{nov}}(q_t) = 1 - \text{SEEN}(q_t)$$

Conceptual illustration of novelty rewards

Avoids global half-space penalties of sigmoid boundaries!



Questioner

Generates questions

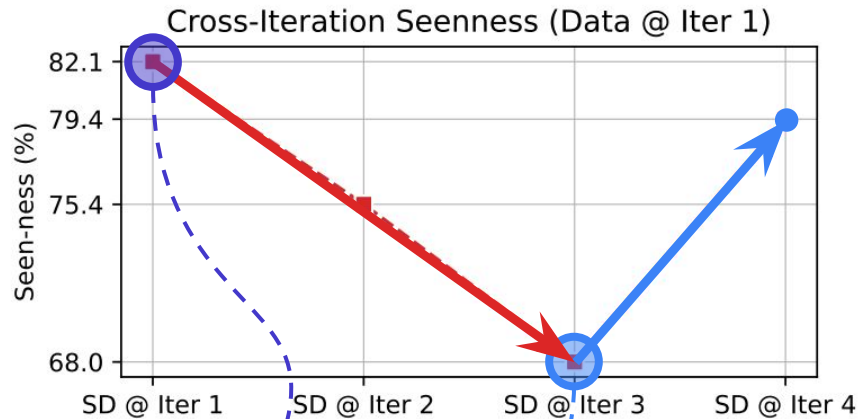
Solver

Generates answers

I've never seen anything like it!

Approach

Re-emergence of Forgotten Questions



Question @ Iter 1

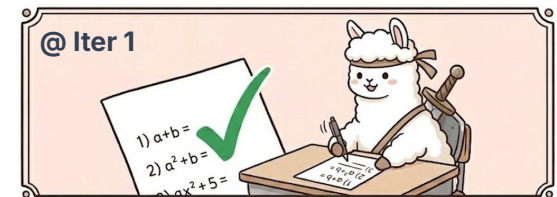
Let a, b, c be the roots of the cubic polynomial $x^3 - 5x^2 + 6x - 1 = 0$.

Find the value of the expression $(\frac{1}{a} + \frac{1}{b} + \frac{1}{c})$.

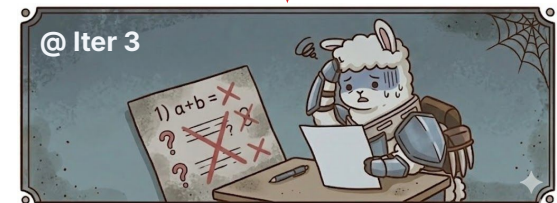
Re-emerged @ Iter 3

Let x_1, x_2, x_3 , and x_4 denote the roots of the equation: $x^4 - 6x^3 + 5x^2 - 4x + 2 = 0$

Determine the value of the sum: $\sum_{k=1}^4 \frac{1}{x_k}$



forgetting questions
as time goes by...



brushing up on
forgotten questions!



Our strategy makes a questioner re-discovers forgotten concepts.

Conclusion

Concluding Remarks

TL;DR

Preventing Iteration Collapse

We prevent iteration collapse in LLM self-play via **One-Class Novelty Reward (OCNR)**.



Thank you!

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Wed, Jul 8, 2026

5:00 PM - 6:45 PM KST

HALL A #2110

for further discussions!

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